NOTE

In publish-subscribe systems, a *topic* is simply a named channel or category that acts as the routing address for messages. Publishers send messages to specific topics without knowing who will receive them, while subscribers express interest in topics without knowing who will send them.

MQTT's publish/subscribe pattern eliminates these dependencies through the abstraction of a topic, where publishers and subscribers do not need to be aware of each other. The MQTT broker acts as an intelligent middleman, routing messages between parties that remain unaware of each other's existence.

For my home digital twin, MQTT provides a unified message bus that can carry data from all sensing sources into my digital twin application. Sensors can be added and removed at any time without impacting consumers of their data.

- LoRaWAN sensors will publish data to MQTT via the LoRaWAN gateway and the AWS IoT Core for LoRaWAN managed network server.
- ESP32 sensors connected by WiFi will publish data directly to MQTT.
- Data from sources other than networked electronic sensors will be captured, transformed, and published to MQTT.

It is important to design a functional topic hierarchy within MQTT to support logical subscription selections and support the future growth of your sensing network. A hierarchical structure that reflects the physical structure of your asset enables you to use wildcards in your subscription patterns. In my home digital twin, I adopt the topic naming convention shown in the following figure, with four topic levels.